5 A small sphere has a charge $q = +1.11 \times 10^{-6}$ C.

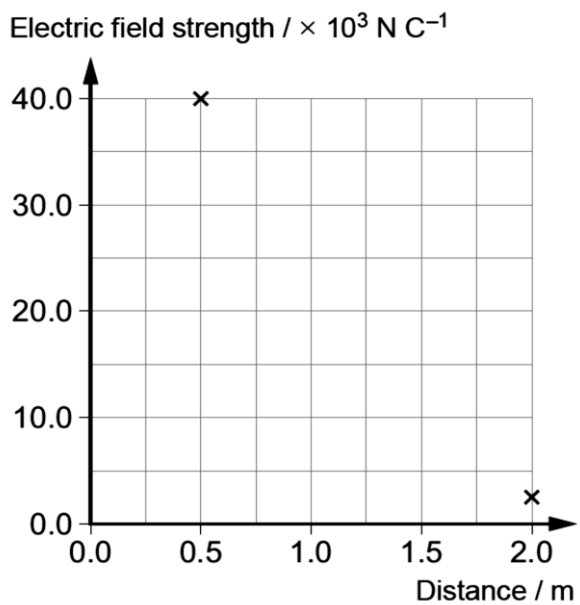
For Examiner's Use

(a) Determine the number of electrons that have been removed from the sphere.

number = ... [2]

(b) On the grids below, sketch curves between 0.5 m and 2.0 m from the centre of the sphere for the electric field strength.

The points at 0.5 m and 2.0 m are already shown.



[2]

(c) A second identical sphere also has a charge of $+1.11 \times 10^{-6}$ C. It is brought from a distant point to a distance 1.2 m from the first sphere as shown in Fig. 5.1.

For Examiner's Use

Fig. 5.1
(not to scale)

(i) Determine the work done required to do this.

work done = …… J [2]

(ii) Determine the magnitude and direction of the electric field strength at the point between the two spheres that is 0.70 m from the left-hand side sphere.

electric field strength = ...…….. N C^{-1}

direction = ……... [3]